

Run 04 | Business exit

India

IV + controls

Target Entrepreneurial/business exit in the past 12 months; binary 1 vs. 0
Sample 27,106 respondents **Coverage** 1 country; 2015–2022
Positive cases 1,402 (5.17%) **Case time** 1m 01s
Train / val. / test 18,434 / 3,252 / 5,420 (68% / 12% / 20%, stratified)
SCT predictors (12): Self-efficacy, Opportunity perception, Fear of failure, Knowing entrepreneurs, Entrepreneurial intention, Media coverage, Business angel, Established business owner, Good career choice, Ease of starting a business, High status, Social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 25.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: CatBoost (0.8434). Highest F1: Random Forest (0.3624). For CatBoost, adding controls changed ROC-AUC by +0.0214 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
CatBoost	0.8434	0.3830	94.15	41.40	31.79	0.3596	18s
H2O GBM	0.8418	0.3827	93.30	35.54	36.43	0.3598	5s
XGBoost	0.8399	0.3784	94.80	49.33	26.43	0.3442	<1s
Random Forest	0.8341	0.3861	94.87	50.64	28.21	0.3624	2s
Elastic Net	0.8123	0.2996	92.14	27.61	32.14	0.2970	7s
Lasso	0.8123	0.2996	92.20	27.86	32.14	0.2985	7s
Ridge	0.8111	0.2926	94.21	39.88	23.93	0.2991	13s
Logistic Regression	0.8109	0.3174	93.10	33.09	32.86	0.3297	<1s
Naive Bayes	0.7605	0.1696	92.88	28.28	24.64	0.2634	7s
Decision Tree	0.6046	0.1937	94.67	45.95	18.21	0.2609	<1s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Business angel	126.11	16.23
2	Entrepreneurial intention	87.72	11.29
3	Age	69.00	8.88
4	Occupational status	56.95	7.33
5	Household size	45.59	5.87
6	Household income	37.47	4.82
7	Education attainment	24.90	3.20
8	Knowing entrepreneurs	24.84	3.20

Share is the percentage of total forest impurity importance across all 64 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Investment environment	9.82	1.26
2	Safety and security	6.76	0.87
3	Social capital	6.70	0.86

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 5 batch, 29 September 2026; 04_obj5_india_iv_cv_EXIT_ENT_harmonized. Scores and importance are taken directly from saved CSV outputs.